

Grothendieck's Galois Theory

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1 Introduction

In this note we give a soft introduction to Grothendieck's interpretation of Galois theory. Galois correspondence is a phenomenon that widely exists in many branches of mathematics. The classical Galois theory of fields gives an inclusion-reversing correspondence between subextensions and subgroups of Galois group of the extension.

In algebraic topology, we also have the following correspondence between covering spaces and subgroups of fundamental group which is surprisingly similar to the classical purely algebraic theorem:

Theorem 1.1 (Classification of covering space)

Let X be a path-connected, locally path-connected, semi-locally simply-connected topological space. Let $\tilde{X}$ be its universal covering. Then:

- (1) There is a one-to-one, contravariant correspondence between subgroups of $G = \pi(X, x)$ and connected coverings of X . In this correspondence, conjugate subgroups correspond to isomorphic coverings and normal subgroups correspond to normal coverings.

(2) The group of covering automorphisms of X^H is isomorphic to $N_G(H)/H$.

It was not until the 1960s, when Alexandre Grothendieck was able to find the proper generalization of a topology on algebraic objects and a corresponding algebraic notion of the fundamental group, that a full algebraic analogue of the topological situation and the direct relation to Galois theory was demonstrated. In Grothendieck's Galois theory, the étale fundamental group is a generalization of the absolute Galois group of a field and the fundamental group in algebraic topology, and the classical Galois theory for fields and classification theorems for covering spaces are just incarnations of this theory. This notes aims to gives a first introduction for Grothendieck's work. We shall discuss Krull's Galois theory for infinite extensions in section 2, and then move to a main theorem that gives an anti-equivalence between finite étale algebras and finite sets equipped with a continuous action of the absolute Galois group in section 3. Some advanced theorems are stated without proof, and the axiomatic approach of Galois category is not treated in this note.

In the sequel we shall assume the basic knowledge of Galois theory for finite extension of fields and properties of separable and normal extensions. Recall that separable extensions of a field k form a distinguished class, in particular, the composition of all separable extensions in a given algebraic closure $\bar{k}$ is separable, this is the splitting field of all separable polynomials over k , and is said to be the **separable closure** k_s of k . Clearly $k_s|k$ is Galois, the Galois group $\text{Gal}(k_s|k)$ is called the **absolute Galois group** of k , and any separable extension of k can be imbedded in k_s .

2 Infinite Galois Extension

This section we explain the infinite Galois extension.

2.1 Profinite Group

A **profinite group** is defined to be an inverse limit of a system of finite groups. In particular, a pro- p group is an inverse limit of a system (G_j, φ_{ij}) of finite p -groups. For algebraic objects, inverse limit can be constructed by the subobject of of the direct product

$$\left\{ (g_j) \in \prod_j G_j \mid g_i = f_{ij}(g_j), i \leq j \right\}$$

Profinite groups are endowed with a natural topology induced by giving G_j the discrete topology thus each φ_{ij} is continuous. It is generated by fibers of φ_j , which coincides with the subspace topology induced by the imbedding $G \hookrightarrow \prod_j G_j$. Note that any connected subspace of G projected to a single point on G_j , thus the only connected subspaces are the single points. It's straightforward to verify that G is a closed subgroup of $\prod_j G_j$. It follows by Tychonoff's theorem that G is a compact and totally disconnected topological group. (In fact the converse is also true).

Lemma 2.1 (Compact Group)

The open subgroups of a compact topological group G are precisely the closed subgroups with finite index

Proof. On the one hand any open subgroup $H \leq G$ is complement of disjoint union of cosets other than H , thus is closed. By compactness, there can only be finite many cosets. On the other hand, any closed subgroup with finite index is the complement of finite many cosets of H , thus open. $\square$

Note that for compact groups, any surjective map is a quotient map, thus:

Lemma 2.2 (First Isomorphism Theorem)

Let G be a compact topological group. An epimorphism of topological group $\varphi : G \rightarrow H$ induces an isomorphism of topological groups $\bar{\varphi} : G/\text{Ker}\varphi \simeq H$

Here are some examples of profinite groups:

2.3 Example (Inverse limit)

(1) Given a group G , the set of its finite quotients can be made into an inverse system: if the normal subgroups $H \leq J$, then there is a canonical projection $G/J \rightarrow G/H$. The inverse limit of the finite quotient groups via the projection is called the profinite completion of G , denoted by $\hat{G}$. In particular, $\hat{\mathbb{Z}} = \varprojlim \mathbb{Z}/n\mathbb{Z}$ with $n \in \mathbb{Z}$ ordered by divisibility.

(2) Consider the inverse system $\mathbb{Z}/p^n\mathbb{Z}$, the inverse limit is called p -adic integers $\mathbb{Z}_p$. By Chinese Remainder Theorem $\mathbb{Z}_n$ is direct sum of $\mathbb{Z}_m$ with m primary factors of n , hence the direct product of $\mathbb{Z}_p$ for all primes is isomorphic to $\hat{\mathbb{Z}}$.

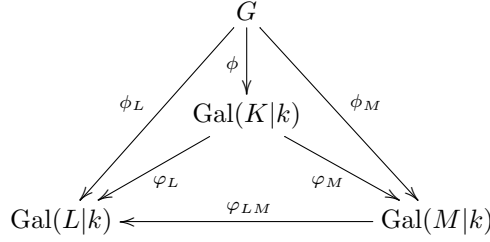
Let $K|k$ be an algebraic Galois extension, for finite Galois subextensions $M|L|k$, there is a natural projection $\varphi_{LM} : \text{Gal}(M|k) \rightarrow \text{Gal}(L|k)$ given by restriction, the map is clearly surjective with kernel $\text{Gal}(M|L)$.

Proposition 2.4 (Krull Topology)

Let $K|k$ be algebraic Galois, then $\text{Gal}(K|k)$ is the inverse limit of Galois group of finite Galois subextensions of $K|k$ along the canonical projection. Thus $\text{Gal}(K|k)$ is a profinite group. A basis of topology (**Krull Topology**) of $\text{Gal}(K|k)$ at identity is given by $\text{Ker}\varphi_{LM} = \text{Gal}(K|L)$.

Proof. Let $\varphi_L : \text{Gal}(K|k) \rightarrow \text{Gal}(L|k), \sigma \mapsto \sigma_L$ be the restriction map. This gives a projective cone over the directed system. Note that $K|k$ is the union of all finite Galois extension. If (G, ϕ_L) is another projective cone, we have a map ϕ from G to $\text{Gal}(K|k)$, defined by gluing together the map ϕ_L s.t. the following diagram commutes. This map is clearly unique so that $(\text{Gal}(K|k), \varphi_L)$

satisfies the universal property of the inverse limit.



Clearly the topology of $\text{Gal}(M|k)$ is generated by fibers of φ_{LM} , and the neighborhood basis at identity should be exactly $\text{Ker} \varphi_L = \text{Gal}(K|L)$. $\square$

2.2 Generalized Fundamental Theorem of Galois

For finite Galois extension, there is a nice bijection between all subgroups of Galois group and intermediate fields. While in infinite case, not all subgroups arise as some Galois group of some subextension.

The term "closed" is always interpreted as topologically closed.

Theorem 2.5 (Generalized Fundamental Theorem)

Let $K|k$ be algebraic Galois and L an intermediate field, then there is a bijection between the closed subgroups of $\text{Gal}(K|k)$ and intermediate fields L , given by

$$L \mapsto \text{Gal}(K|L) \text{ and } H \mapsto K^H$$

such that:

- (1) $L|k$ is finite iff $\text{Gal}(K|L)$ is open, in which case $[L : k] = [\text{Gal}(K|k) : \text{Gal}(K|L)]$.
- (2) There is an equivariant action of $\text{Gal}(K|k)$ given by conjugation on closed subgroups and by translation on intermediate fields.
- (3) $L|k$ is Galois iff $\text{Gal}(K|L)$ is normal in $\text{Gal}(K|k)$, in which case $\text{Gal}(L|k) \cong \text{Gal}(K|k)/\text{Gal}(K|L)$ as topological groups.

Proof. First we show the correspondence is well defined. Write $L|k$ as a union of finite subextensions $L_j|k$. Let N_j be the normal closure of L_j in K . Now $\text{Gal}(K|L)$ is precisely the inverse image of $\text{Gal}(N_j|L)$ under the projection $\text{Gal}(K|k) \rightarrow \text{Gal}(N_j|k)$, thus must be open, which implies it's also closed by lemma 2.1. But $\text{Gal}(K|L) = \bigcap_j \text{Gal}(K|L_j)$, thus $\text{Gal}(K|L)$ is closed.

Clearly for any intermediate field L , $K|L$ is Galois by the property of normal and separable extension, thus the map $L \mapsto \text{Gal}(K|L)$ is injective. To show surjectivity, it suffices to show that any closed subgroup H is the Galois group of extension $K|K^H$. Clearly $H \leq \text{Gal}(K|K^H)$. To show equality, let $\sigma \in \text{Gal}(K|K^H)$, and let $U = \text{Gal}(K|N)$ be an open neighborhood about identity in $\text{Gal}(K|K^H)$, with $N|K^H$ some finite Galois subextension. Clearly H projects onto $\text{Gal}(N|K^H)$ via the canonical projection. But this implies there is some $\eta \in H$ s.t. $\eta|_N = \sigma|_N$, i.e. $\eta \in \sigma U$. Since U is arbitrary, we have showed that σ is in the closure of H . But H is already closed, so

$\sigma \in H$.

(1) follows directly by lemma 2.1 and a similar index relationship as the finite case.

For (2) it's easy to check $\sigma \text{Gal}(K|L)\sigma^{-1} = \text{Gal}(K|\sigma L)$ for any $\sigma \in \text{Gal}(K|k)$.

The proof of the last statement is similar to the finite case: Note that $\sigma \text{Gal}(K|L)\sigma^{-1} = \text{Gal}(K|\sigma L)$ and injectivity of $L \mapsto \text{Gal}(K|L)$ implies $\text{Gal}(K|L)$ is normal iff $\sigma L = L$ for any $\sigma \in \text{Gal}(K|k)$. If $L|k$ is Galois, then it's normal and $\sigma L = L$. Conversely, any k -morphism $\sigma_0 : L \rightarrow K$ is extendible to an element $\sigma \in \text{Gal}(K|k)$, thus $\sigma L = L$ and σ_0 has image in L , hence $L|k$ is normal and thus Galois. Let $\varphi : \text{Gal}(K|k) \rightarrow \text{Gal}(L|k)$ be the restriction, we have $\text{Gal}(L|k) \cong \text{Gal}(K|k)/\text{Ker}\varphi = \text{Gal}(K|k)/\text{Gal}(K|L)$ by lemma 2.2.

□

For instance, consider a finite field F with separable closure F_s , there is a unique subextension F_n in F_s of order n , besides, $\text{Gal}(F_n|F) = \mathbb{Z}/n\mathbb{Z}$. Thus, $\text{Gal}(F_s|F) \simeq \hat{\mathbb{Z}}$. But the subgroup $\mathbb{Z} \subset \hat{\mathbb{Z}}$ is not closed.

The absolute Galois group has many remarkable applications in algebraic number theory and algebraic geometry. A famous example is:

Theorem 2.6 (Florian Pop)

Let K, L be two infinite fields that are finitely generated over their prime field. Fix separable closures K_s, L_s of K and L , respectively, and assume there exists a continuous isomorphism $\Phi : \text{Gal}(K_s|K) \simeq \text{Gal}(L_s|L)$ of profinite groups. Then there exist purely inseparable extensions $K'|K, L'|L$ with $K' \simeq L'$. Moreover, there is an isomorphism $\phi : L'_s \simeq K'_s$ of separable closures such that $\Phi(g) = \phi^{-1}g\phi$ for all $g \in \text{Gal}(K'_s|K')$.

3 Finite Étale Algebra

3.1 The category of Galois sets

Given a group G , a G -set is just a set with an action of G . G -morphisms are maps that commute with the action of the group.

Let $L|k$ be finite separable, and k_s be the separable closure in a given algebraic closure $\bar{k}$ (Here we do not consider L as subfield of k_s). Recall that the separable degree $|\text{Hom}_k(L, \bar{k})| = [L : k] < \infty$, and any morphism $L \rightarrow \bar{k}$ must have image in k_s . Thus we may consider the finite set $\text{Hom}_k(L, k_s)$. We shall denote the absolute Galois group of k by G , then there is a left action of G by composition: $(g, f) \mapsto g \circ f$ for $g \in G, f \in \text{Hom}_k(L, k_s)$. The property of this action is summarized in the following lemma:

Lemma 3.1

The above action of G on $\text{Hom}_k(L, k_s)$ is transitive and continuous, thus $\text{Hom}_k(L, k_s)$ as a G -set is isomorphic to set of left coset of an open subgroup of G . In particular, if $L|k$ is

finite Galois, the set of cosets is in fact a quotient of G .

Proof. For any $f_1, f_2 \in \text{Hom}_k(L, k_s)$, let $\varphi : f_1(L) \rightarrow L$ be the inverse of f_1 , then $f_2 \circ \varphi : f_1(L) \rightarrow k_s$ can be extended to some $g \in G$ by normality of $k_s|k$. Clearly $g \circ f_1 = f_2$, thus the action is transitive. The inverse image of $f \in \text{Hom}_k(L, k_s)$ under multiplication is the disjoint union $\bigcup_{h \in \text{Hom}_k(L, k_s)} \{g : gh = f\} \times \{h\} = \bigcup_{h \in \text{Hom}_k(L, k_s)} g_h G_f \times \{h\}$, where g_h is chosen so that $g_h \circ h = f$, and G_f is the stablizer of f . But $G_f = \text{Gal}(k_s|fL)$, which is open by (1) of theorem 2.5. Fixing f , consider the assignment $h \mapsto \{g : gh = f\} = g_h G_f$ it's bijective, and preserves the action of G since for any $a \in G$, $a \{g : gh = f\} = \{g : gh = af\}$. $\square$

Given two finite separable extensions of $\varphi : M, L$ and a k -morphism $\varphi : M \rightarrow L$, we obtain induced morphism $\varphi^* : \text{Hom}_k(L, k_s) \rightarrow \text{Hom}_k(M, k_s)$ by pull-back $f \mapsto f \circ \varphi$. Note that φ^* preserves the action of G . Thus $\text{Hom}_k(-, k_s)$ is a contravariant functor from the category FSE_k of finite separable extensions of k to the category $G\text{-TSet}$ of finite sets with continuous transitive G -action. In fact, we have

Theorem 3.2

The functor $\text{Hom}_k(-, k_s)$ gives an anti-equivalence between FSE_k and $G\text{-TSet}$

Proof. It suffices to prove $\text{Hom}_k(-, k_s)$ is essentially surjective and fully faithful. Given a continuous transitive G -set S , we fix a point $s \in S$, the stablizer of s is an open subgroup G_s of G . Now if G_s has fixed field L , we have a natural action of G on $\text{Hom}_k(L, k_s)$, with stablizer of $L \hookrightarrow k_s$ be $\text{Gal}(k_s|L) = G_s$. Consider the assignment $\text{Hom}_k(L, k_s) \rightarrow S, f \mapsto \tilde{f}s$ (In particular, $L \hookrightarrow k_s$ sent to s), where $\tilde{f} \in G$ extends f . Both G -sets are isomorphic to the left coset $G_s \backslash G$.

For fully faithfulness, let L, M be two finite separable extensions of k . A G -morphism $\phi : \text{Hom}_k(M, k_s) \rightarrow \text{Hom}_k(L, k_s)$ is determined by its restriction on a single point $f \in \text{Hom}_k(M, k_s)$ by transitivity. Note that the stablizer G_f must also fix $G_{\phi(f)}$, i.e. $G_f \leq G_{\phi(f)}$. This means the fixed field $(\phi f)(L) \subset f(M)$. Let f^{-1} be the inverse of $M \rightarrow f(M)$, we have $f^{-1}\phi(f)L \subset M$. Thus, the map $f^{-1}\phi(f) : L \rightarrow M$ is the only map that induces ϕ .

Finally, if $L|k$ is finite Galois, the G -set is isomorphic to a quotient of G by the previous theorem. $\square$

3.2 Finite Étale Algebra

A k -algebra A is **étale** if it's a finite direct product of finite separable extension of k .

Theorem 3.3 (Main Theorem-Grothendieck's Version)

The functor $\text{Hom}(-, k_s)$ gives an anti-equivalence between the category FEAlg_k of finite étale k -algebras and the category $G\text{-Set}$ of finite sets with continuous left G -action. Here separable field extensions give rise to sets with transitive G -action and Galois extensions to G -sets isomorphic to finite quotients of G .

Proof. Suppose $A = \prod_i L_i$, where each L_i is finite separable over k . For $\phi \in \text{Hom}_k(A, k_s)$, it must be a composition of the form $A \twoheadrightarrow L_i \rightarrow k_s$ for some i , in other words, $\text{Hom}_k(A, k_s)$ is a disjoint union of $\text{Hom}_k(L_i, k_s)$. The action of G is just given by the action on each orbit $\text{Hom}_k(L_i, k_s)$. If $A' = \prod_j L'_j$ is another étale algebra, a morphism $A \rightarrow A'$ is just a collection of morphisms $\{L_i \rightarrow L'_j\}_i$. By the previous theorem, we have a bijective correspondence between the algebra morphism and G -morphisms of sets. $\square$

We can generalize the theorem to arbitrary Galois extensions.

Theorem 3.4

Let $K|k$ be algebraic Galois, then the functor $\text{Hom}(-, K)$ gives an anti-equivalence between the category of finite étale k -algebras that are products of subfields of K and the category of finite sets with continuous left $\text{Gal}(K|k)$ -action. Here separable field extensions give rise to sets with transitive $\text{Gal}(K|k)$ -action and Galois extensions to $\text{Gal}(K|k)$ -sets isomorphic to finite quotients of $\text{Gal}(K|k)$.

3.3 Galois Theory for Schemes

We end the note by stating the general theorem on Galois theory for schemes without proof. The previous main theorem 3.3 for field extensions is actually a prototype of this result.

A morphism $f : Y \rightarrow X$ of schemes is finite étale if there exists a covering of X by open affine subsets $U_i = \text{Spec} A_i$, such that for each i the open subscheme $f^{-1}(U_i)$ of Y is affine, and equal to $\text{Spec} B_i$, where B_i is a free separable A_i -algebra. In this situation we also say that $f : Y \rightarrow X$ is a **finite étale covering** of X .

Theorem 3.5 (Main Theorem for Schemes)

Let X be a connected scheme. Then there exists a profinite group G , uniquely determined up to isomorphism, such that the category of finite étale coverings of X is equivalent to the category of finite sets on which G acts continuously.

The profinite group G occurring in the theorem is called the fundamental group of X denoted by $\pi(X)$.

Thus, theorem 3.3 is a special case of this result, with $X = \text{Spec} k$. Moreover, we have the following analogy for topological spaces:

Theorem 3.6

Let X be a connected topological space. Then there exists a profinite group $\hat{\pi}(X)$, uniquely determined up to isomorphism, such that the category of finite coverings of X is equivalent to the category $\hat{\pi}(X)$ -sets of finite sets on which $\hat{\pi}(X)$ acts continuously.

If X is a non-singular variety over $\mathbb{C}$, and X_h is the associated complex analytic space, then the algebraically defined fundamental group $\pi(X)$ from theorem 3.5 is isomorphic to $\hat{\pi}(X_h)$, which

is in fact the profinite completion of the classical fundamental group.

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